

Lema de reducción a la parte real para aplicaciones lineales complejas

Lema 1. Sea X un $\mathbb{C}$ -espacio vectorial y sea $L: X \rightarrow \mathbb{C}$ una aplicación lineal

$$\implies \forall x \in X : L(x) = \Re(L(x)) - i \Re(L(ix)).$$

Demostración: Sea $x \in X$. Entonces, $L(x) = \Re(L(x)) + i \Im(L(x))$. Pero, por linealidad,

$$L(ix) = iL(x) = i\Re(L(x)) - \Im(L(x)) \implies \Re(L(ix)) = -\Im(L(x)).$$

Por tanto, $\Im(L(x)) = -\Re(L(ix))$ y $L(x) = \Re(L(x)) - i \Re(L(ix))$. ■

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